M4C 2025 - Assignment 3

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Problem 1

(a) Euler's totient function $\phi(n)$ is the function that maps a natural number $n \in \mathbb{N}$ to its number of coprimes that are smaller than n. Because $n = pq$, $\phi(n)$ is the number of coprimes of pq from 1 up to pq.

(b) Remark: I got a bit carried away with exercising my formal proving ability which means this question was answered more thoroughly than it was probably expected (which is becoming my trait at this point).

Let's first prove the next statement: *Intergers smaller than $n = pq$ are coprime to p and q iff they are coprime to n.*

This can be explained by the definition of coprime numbers. Let's first prove the implication to the right. Let $m < n$ be a natural number that is coprime to p and q. Two numbers m and n are coprime iff $\gcd(m, n) = 1$. Because $n = pq$, the common divisers of m and n can only be: $1, p, q, n$ by the definition of prime numbers. From this we can learn that $p|m \implies \gcd(m, n) \neq 1$ and symmetrically $q|m \implies \gcd(m, n) \neq 1$. Because m is by assumption coprime to p and q, it follows that $\gcd(m, n)$ can only be 1 or n. But because $m < n$, it can only be 1.

The implication to the left is easier. If we know that $\gcd(m, n) = 1$, we know that $p \not| m$ and $q \not| m$. Because p and q are prime, it naturally follows m is coprime to both p and q. $\square$

From the statement above we can learn that the only two numbers that are not coprime to n and are still smaller than n are exactly p and q (so exactly 2 integers). If we don't limit ourselves to smaller coprime numbers, there are infinitely amount of coprimes to n (which would be divisible by at least one of p, q or n).

(c) Because p and q are primes, we know that all smaller numbers than them are their coprimes, so: $\varphi(p) = (p - 1)$ and $\varphi(q) = (q - 1)$. Because we proved that the coprimes to n are exactly the primes to p and the primes to q, we can derive the value of the totient function:

$$\varphi(pq) = \varphi(p)\varphi(q) = (p - 1)(q - 1)$$

Problem 2

(a) ...

Problem 3

(a) From the definition of our elliptic curve we can learn that $A = 2$ and $B = 4$. We need to check that $4 * 2^3 + 27 * 4^2 \not\equiv 0 (\mod 7)$. We can first change 27 into 6 ($\mod 7$) from where we can easier calculate:

$$4 * 2^3 + 6 * 4^2 = 2 (\mod 7) \not\equiv 0 (\mod 7)$$

(b)

b) MODULO 7 ; $\boxed{y^2 = x^3 + 2x + 4}$

$x = 0 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2$

$x = 1 \not\Rightarrow y^2 = 7$

$x = 2 \not\Rightarrow y^2 = 2$

$x = 3 \not\Rightarrow y^2 = 2$

$x = -1 = 6 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1$

$x = -2 = 5 \not\Rightarrow y^2 = 6$

$x = -3 = 4 \not\Rightarrow y^2 = 6$

$y = 0 \rightsquigarrow$ No solution

$y = 1 \rightsquigarrow y^2 = 1$

$y = 2 \rightsquigarrow y^2 = 4$

$y = 3 \rightsquigarrow y^2 = 2$

$y = -1 \rightsquigarrow y^2 = 1$

$y = -2 \rightsquigarrow y^2 = 4$

$y = -3 \rightsquigarrow y^2 = 2$

$E(\mathbb{F}_7) = \{(0,2), (0,5), (2,3), (2,4), (3,3), (3,4), (6,1), (6,6)\}$

(c)

c) $\underline{\text{ord}(2,3) = 5} \iff \underline{\underline{[5](2,3) = \infty}}$

$[2](\overset{x_0\ y_0}{2,3}) = (3,4)$

$\lambda_2 = \dfrac{3x_0^2 + A}{2y_0} = \dfrac{3 \cdot 2^2 + 2}{2 \cdot 3} = \dfrac{14}{6} = 0$

$x_2 = \lambda_2^2 - 2x_0 = -4 \equiv 3 \pmod 7$

$y_2 = -y_0 + \lambda_2(x_0 - x_2) = -3 = 4 \pmod 7$

$[4](2,3) = \underline{[2](3,4)} = (2,4)$

$\lambda_4 = \dfrac{3 \cdot 3^2 + 2}{2 \cdot 4} = 1 \cdot 1^{-1} = 1$

$x_4 = 1 - 2 \cdot 3 = 2 \pmod 7$

$y_4 = -4 + (3 - 2) = 4 \pmod 7$

$[5](2,3) = [4](2,3) + (2,3) = \underline{(2,4) + (2,3)} = \infty$

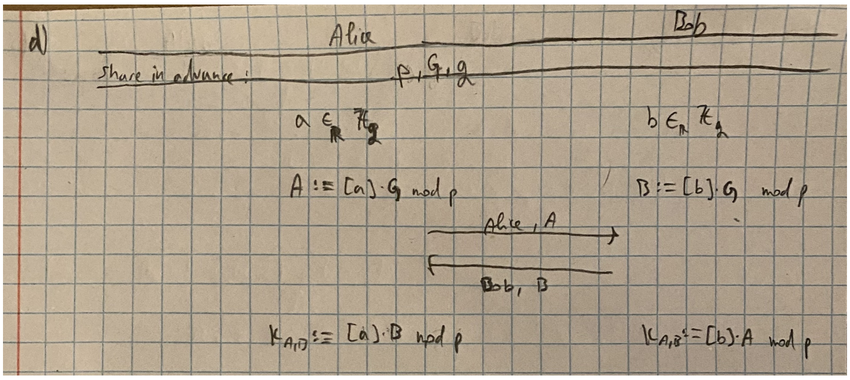
$\lambda_5 = \dfrac{y_0 - y_4}{x_0 - x_4} = \dfrac{4-3}{2-2} = \dfrac{1}{0} = \infty$

$x_5 = \lambda^2 - x_4 - x_0 = \infty - 4 - 3 = \infty$

$y_5 = -4 + \infty = \infty$ ∎

Remark: I have later noticed that at the last step I didn't need to compute λ since we could use the fact that $P = -Q$ is true. Calculating the λ then of course produces the point at infinity because the line going through P and Q is vertical.

(d) The main thing that changes from regular DH to ECDH is that we swap multiplication for addition which means that exponentiation becomes multiplication:



(e) We have a base/starting point $G = (2, 3)$. Alice has chosen a secret value $a = 2$ from which she generates $A := [a]G = [2](2, 3)$. Bob chose a secret value $b = 3$ from which he generates $B := [b]G = [3](2, 3)$. Alice and Bob then publically exchange A and B. This can be done without reveiling either a or b because of the hardness of discrete log problem (we cannot efficiently calculate a from $[a]G$ just like we couldn't calculate a from g^a in DH). Alice then calculates $K_{A,B} := [a]B \mod p = [2][3](2, 3) = [5](2, 3) + (2, 3) = 0 + (2, 3) = (2, 3)$ and Bob symmetrically computes $K_{A,B} := [b]A \mod p = [3][2](2, 3) = [2][3](2, 3)$ due to commutativity.